MALE REPRODUCTIVE SYSTEM

- Organs
- Functions
- Pathway of Sperm Cells

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B S Accountancy 2

OBJECTIVES

1. Identify the structures and major organs of the male reproductive system and describe their functions.
2. Trace the pathway of the sperm cells in the organ system.

What is the
MALE REPRODUCTIVE SYSTEM?

Male Reproductive System

- Consists of a number sex organs that are a part of the human reproductive process.

- Produces, stores and releases the male gametes, or sperm.

**gametes*- a sex cell

MALE REPRODUCTIVE ORGANS

External Genital Organs

1. Penis
2. Scrotum

Internal Genital Organs

1. Testis
2. Epididymis
3. Vas Deferens
4. Accessory Glands
   a. Seminal Vesicles
   b. Prostate Gland
   c. Bulbourethral Glands

External Genital Organs

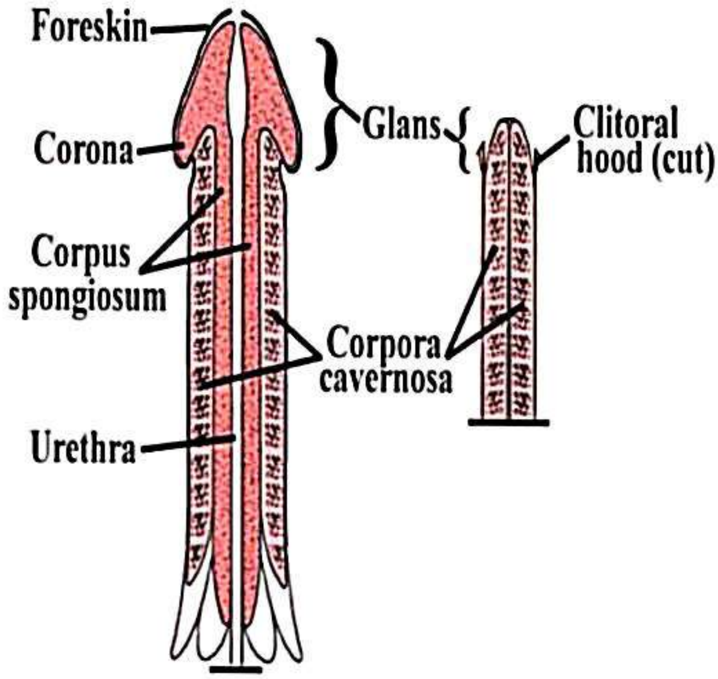
Penis

Scrotum

Penis

The **penis** is the organ by which the sperm is introduced into the female.

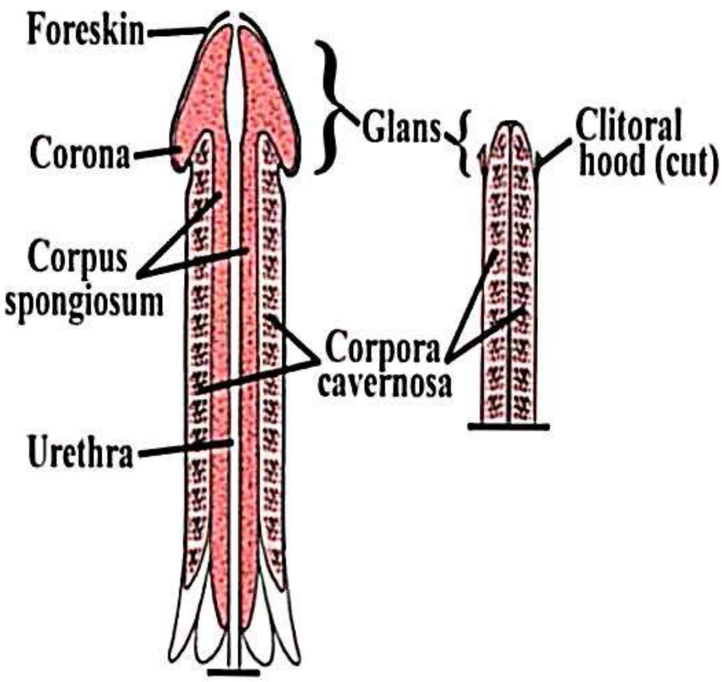
It contains spongy tissue that becomes turgid and erect when filled with blood.



Cont...

- **Erectile Tissues**
  - ***Corpus spongiosum*** – is the mass of spongy tissue which surrounds urethra and involves in erection by allowing rushing of blood into it

  - ***Corpus cavernosa*** – is one of a pair of songe-like regions of erectile tissue which contains most of the blood in the penis during penile erection

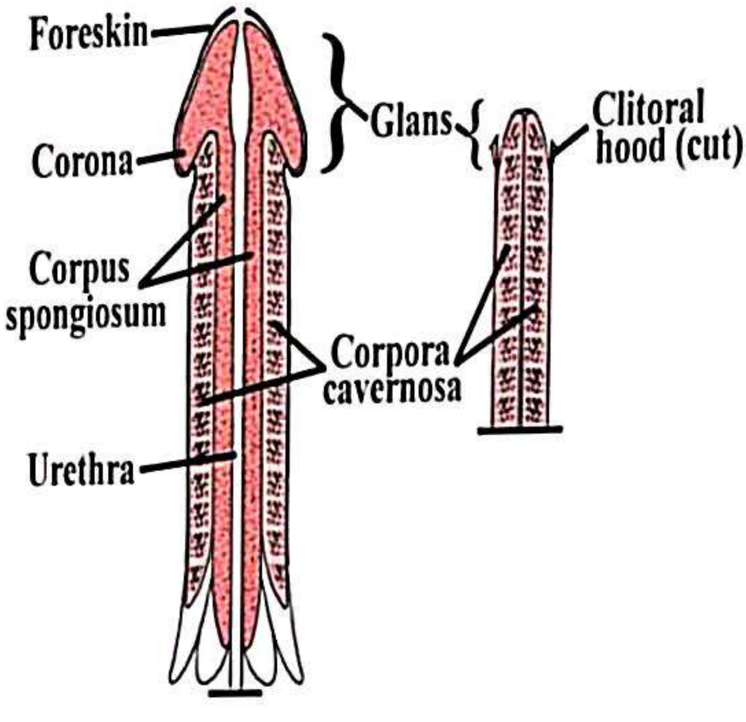


Cont...

- **Urethra** – a tube within the penis that conveys semen out of the body during ejaculation.

- **Glans** – the rounded, highly sensitive head of the penis.

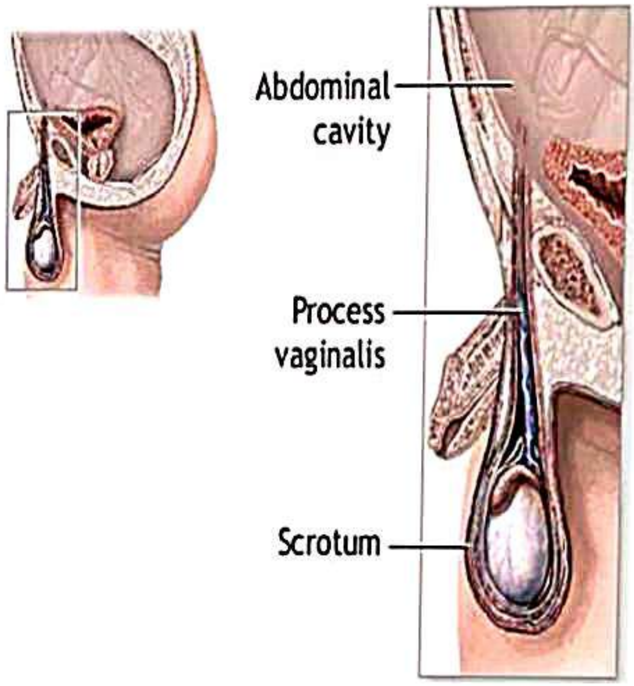
- **Prepuce** – a fold of skin, covering the head of the penis.



Scrotum

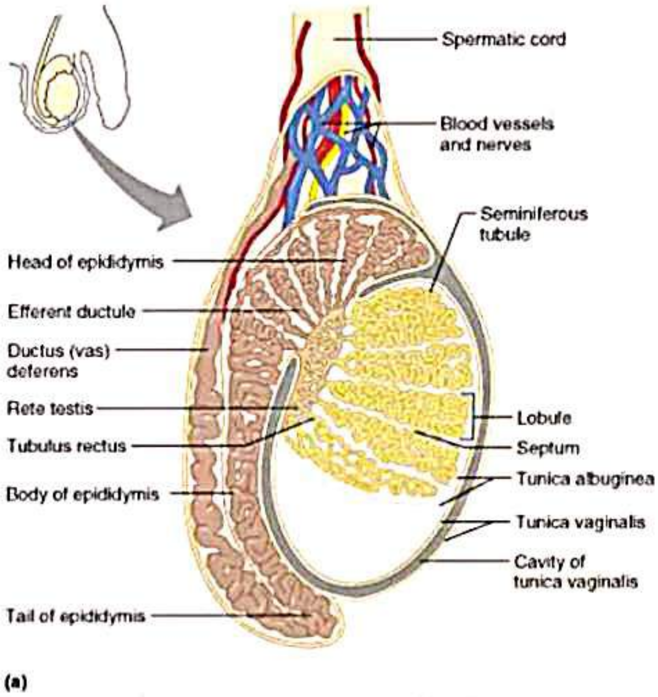
A pouch of skin formed from the lower part of the abdominal wall.

The **scrotum** keeps the testes at a temperature slightly cooler than body temperature.



Testis *(plural* testes)

The testes are the two-oval shaped male organs that produce sperm and hormone testosterone.



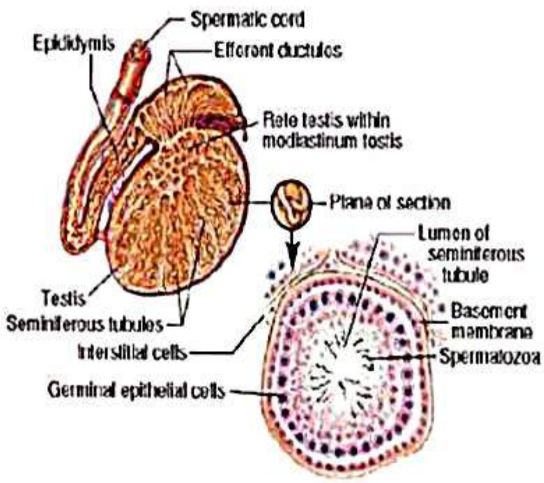
****Testosterone***- the primary male sex hormone

Cont…

Each testis is made of tightly coiled structures called ***seminiferous tubules.***

Among tubules are cells that produce testosterone.

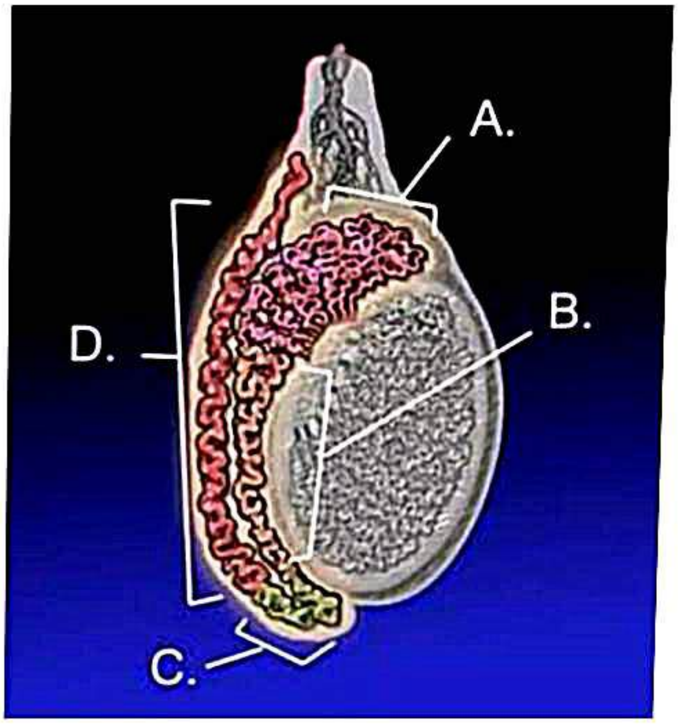
Seminiferous Tubules



Epididymis

The epididymis is a tightly coiled tubes against the testicles.

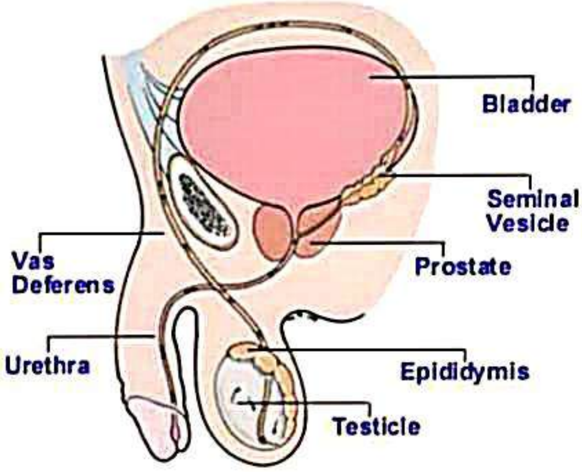
It acts as maturation and storage place for sperm.



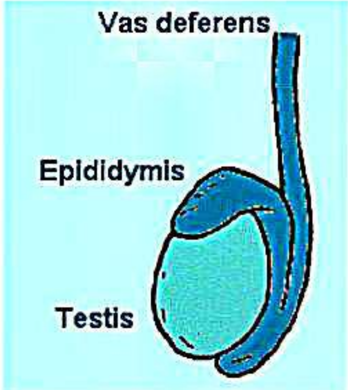
Adult human testicle with epididymis:
A. Head of epididymis,
B. Body of epididymis,
C. Tail of epididymis, and
D. Vas deferens

Vas Deferens (Ductus Deferens)

The vas deferens is a thin tube that starts from the epididymis to the urethra in the penis.



They transport sperm from the epididymis in anticipation of ejaculation.



Accessory glands

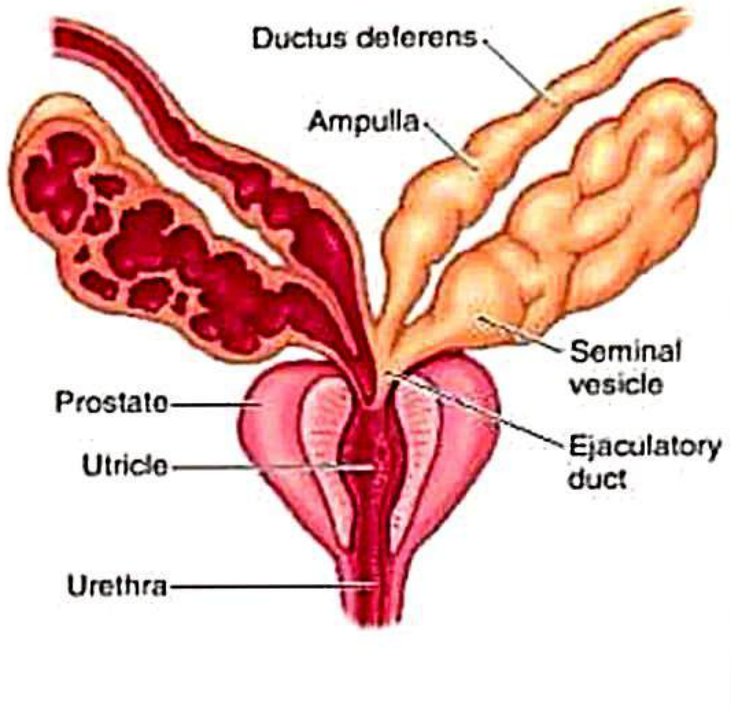
a. **Seminal Vesicles**
b. **Prostate Gland**
c. **Bulbourethral Glands**

These glands produce nourishing fluids for the sperms that enter the urethra.

Seminal Vesicles

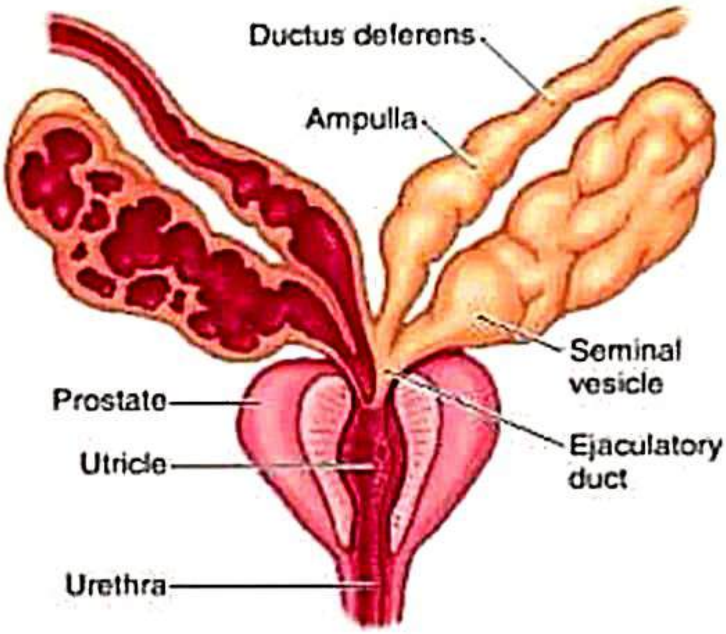
The Seminal Vesicles are sac-like structures attached to the vas deferens at one side of the bladder.

They produce a sticky yellowish fluid that contains fructose.



Prostate Gland

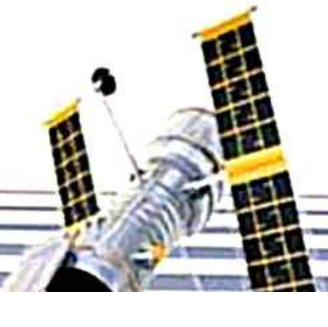
The Prostate Gland surrounds the ejaculatory ducts at the base of the urethra, just below the bladder.



The Prostate Gland is responsible for making the production of semen, a liquid mixture of sperm cells, prostate fluid and seminal fluid.

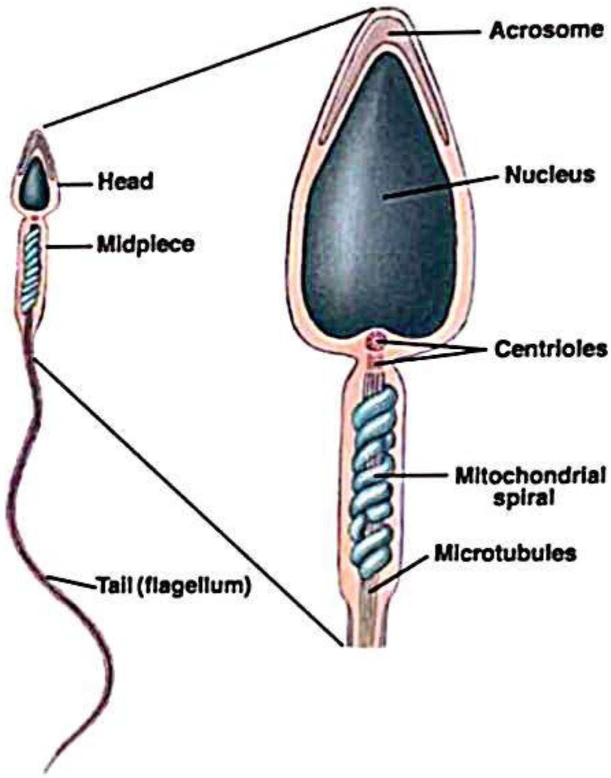
REVIEW:

- The main function of the Male Reproductive System is to produce sperm cells and deliver them to the female reproductive system.
- It consists of external and internal genital organs which are essential for the continuous reproduction of life.



SPERM

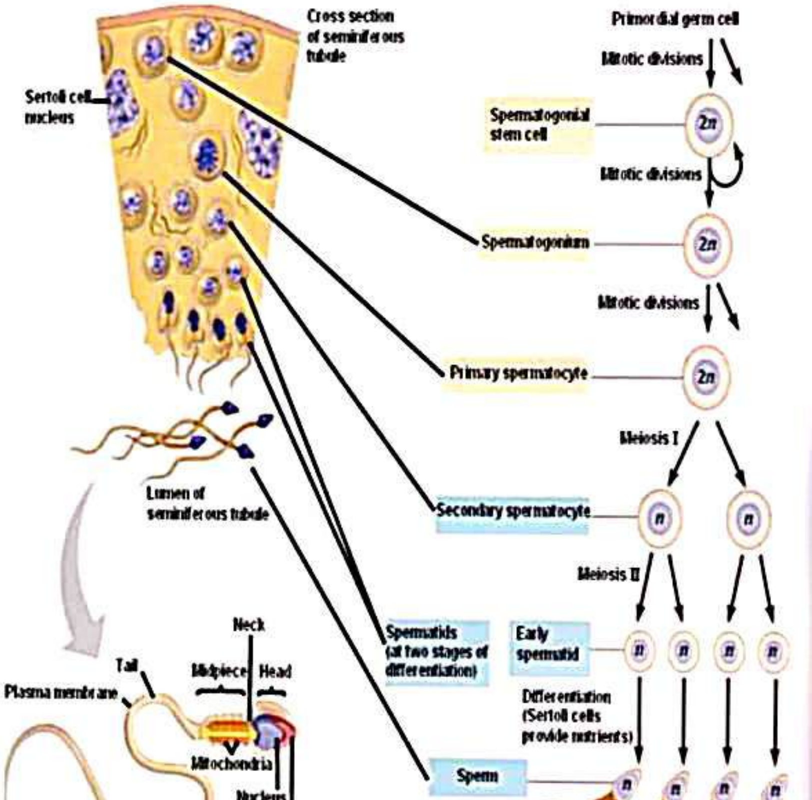
- *Function:*
  - To move and carry genetic information to the egg.
- *Structure:*
  - **Head**: The large head region of the sperm that contains DNA.
  - **Midpiece**: The narrow middle part of the cell that contains mitochondria.
  - **Tail**: The wavelike motion of the flagellum propels the sperm forward.



SPERMATOGENESIS

Spermatogenesis is the formation of sperm cells.

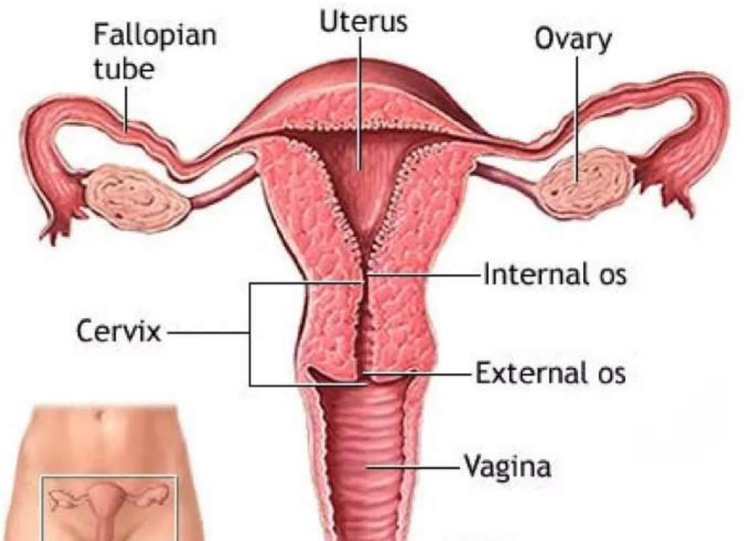
It takes place in the seminiferous tubules.



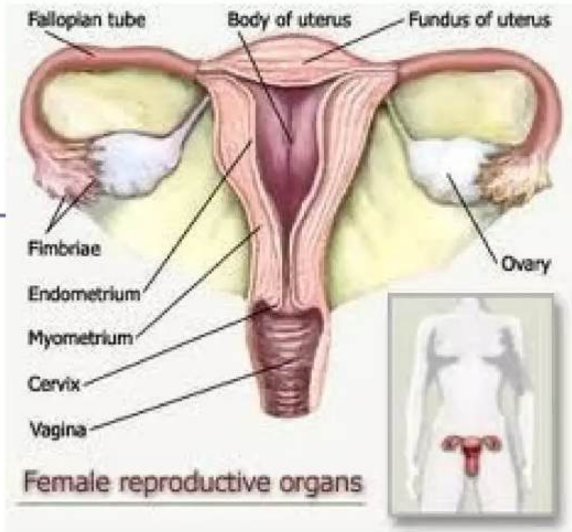
Female Reproductive System

- Produce sex hormones
  - Estrogen, Progesterone
- Produce egg (ova)
- Support & protect developing embryo
- Give birth to new baby

Female Reproductive System



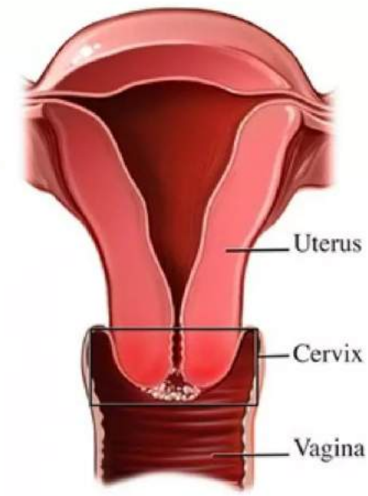
Major Organs



- Cervix
- Vagina
- Ovaries [gonads]
- Uterine tubes [fallopian tubes]
- Uterus

The Cervix

- The lower portion or neck of the uterus.
- The cervix is lined with mucus, known as cervical mucus
  - Cervical mucus provides lubrication & sperm transport during sexual intercourse
  - During ovulation secretion of cervical mucus increases in response to estrogen
  - But when an egg is ready for fertilization, the mucus then becomes thin and slippery, offering a “friendly environment” to sperm

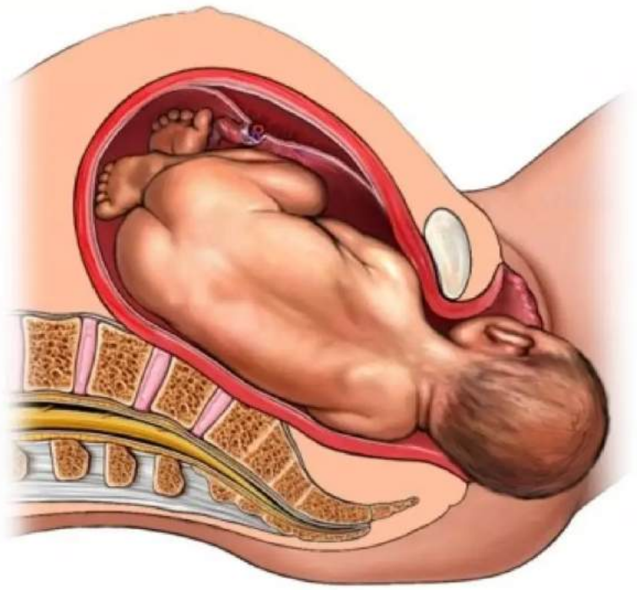


The Cervix

At the end of pregnancy

The cervix acts as the passage through which the baby exits the uterus into the vagina.

The cervical canal expands to roughly 50 times its normal width for the passage of the baby during birth

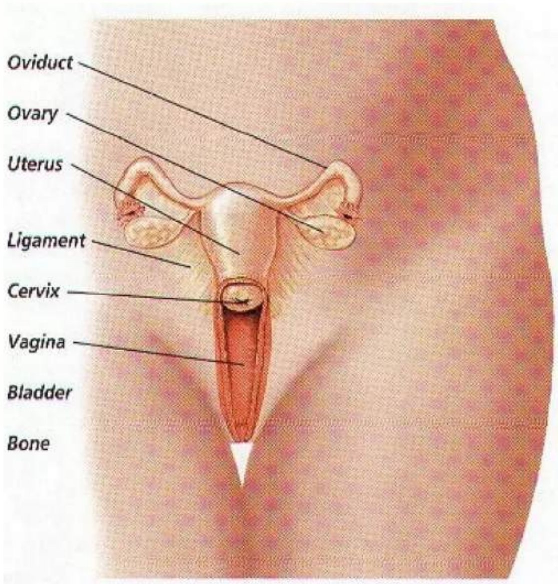


The Vagina

- A muscular, ridged sheath connecting the external genitals to the uterus.

- Functions as a two-way street, accepting the penis and sperm during intercourse

- Serving as the avenue of birth through which the new baby enters the world

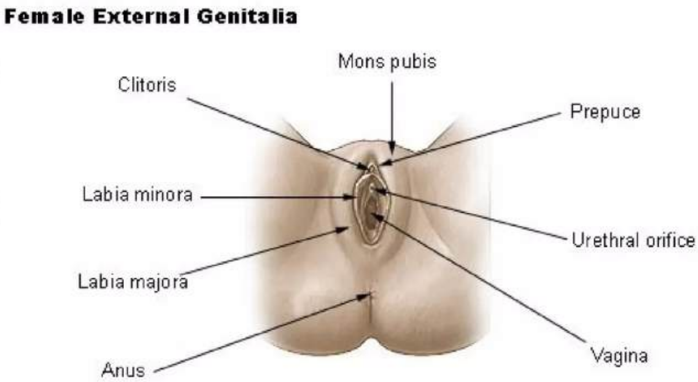


External genetalia

- **Vulva**—which runs from the pubic area downward to the rectum.

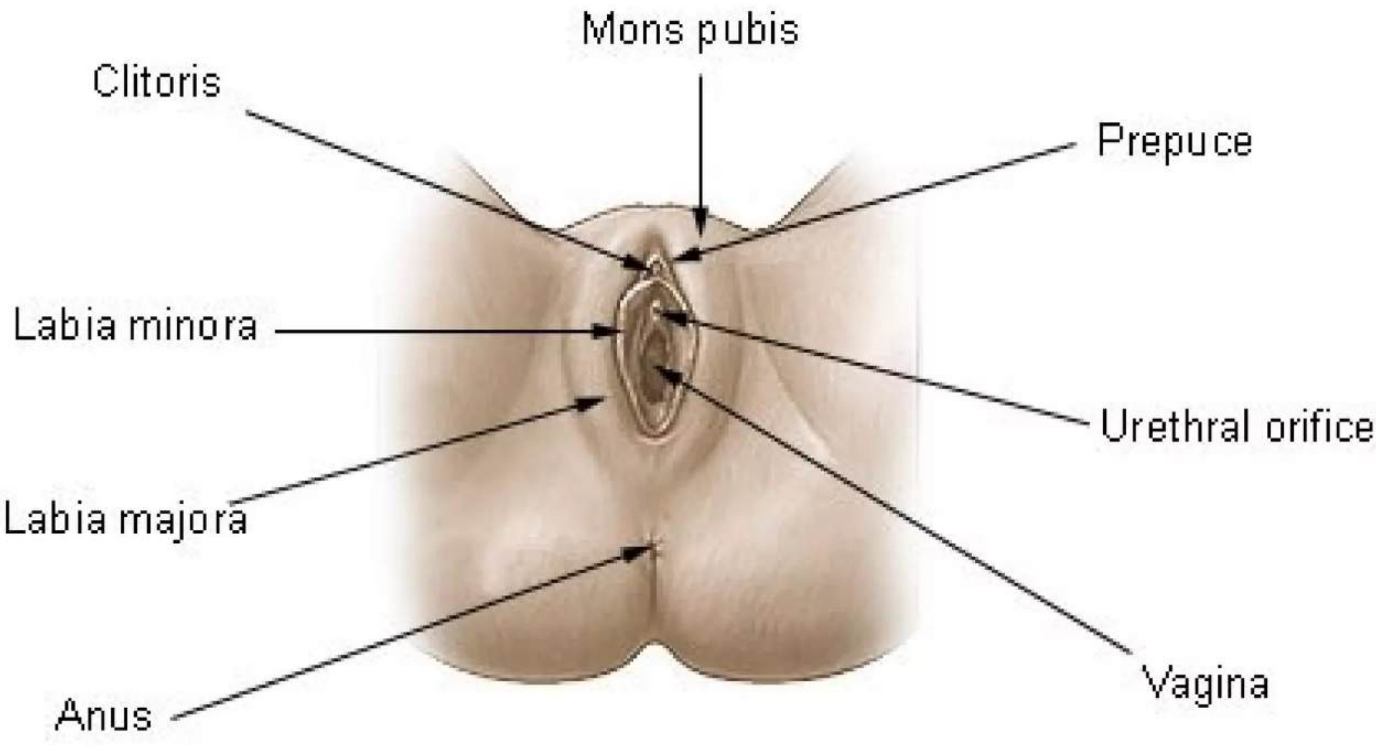
- **Labia majora** or "greater lips" are the part around the vagina containing two glands (Bartholin’s glands) which helps lubrication during intercourse.

- **Labia minora** or "lesser lips" are the thin hairless ridges at the entrance of the vagina, which joins behind and in front. In front they split to enclose the clitoris



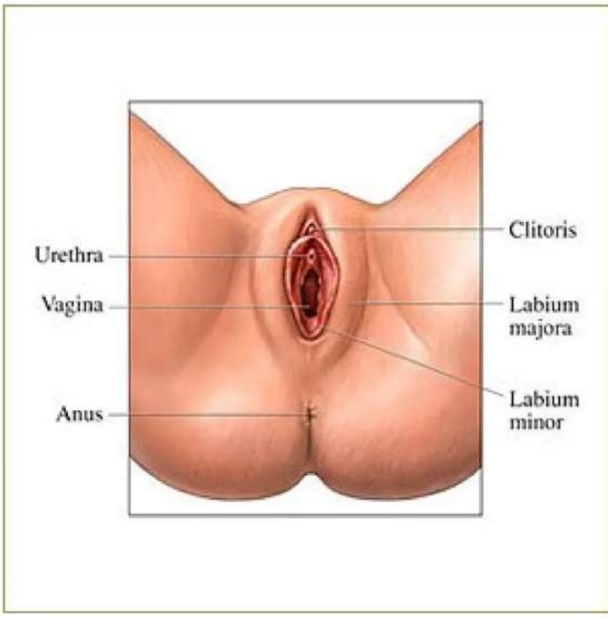
- **The clitoris** is a small pea-shaped structure. It plays an important part in sexual excitement in females.

Female External Genitalia



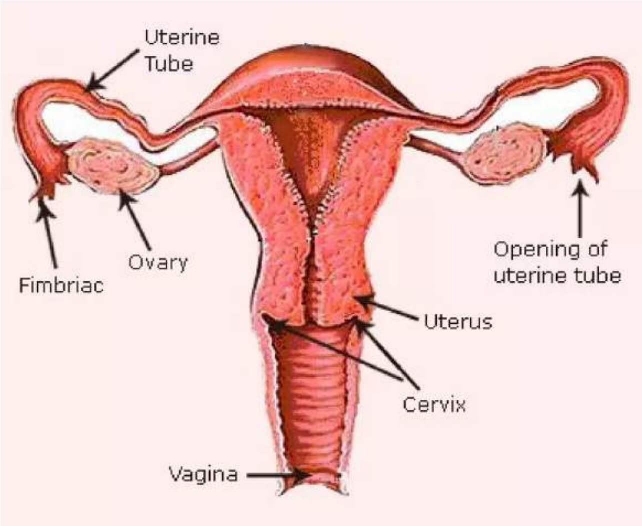
External genetalia

- The urethral orifice or external urinary opening is below the clitoris on the upper wall of the vagina and is the passage for urine
- Opening of the vagina is separate from the urinary opening and located below it.
- The hymen is a thin cresentic fold of tissue which partially covers the opening of the vagina. medically it is no longer considered to be a 100% proof of female virginity.

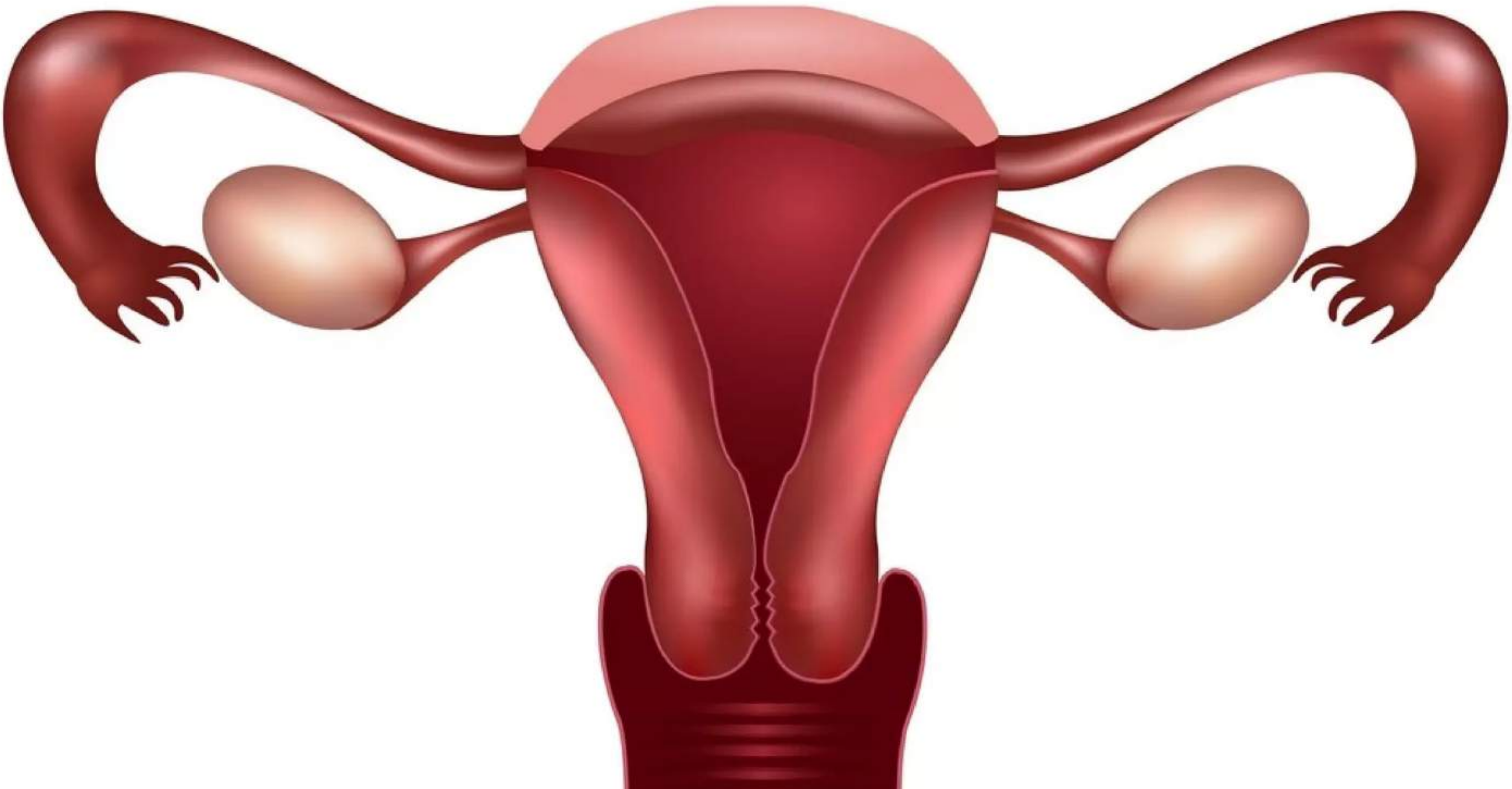


Ovaries

- Also known as female gonads
- They produce eggs (also called ova) every female is born with a lifetime supply of eggs
- They also produce hormones: **Estrogen & Progesterone**

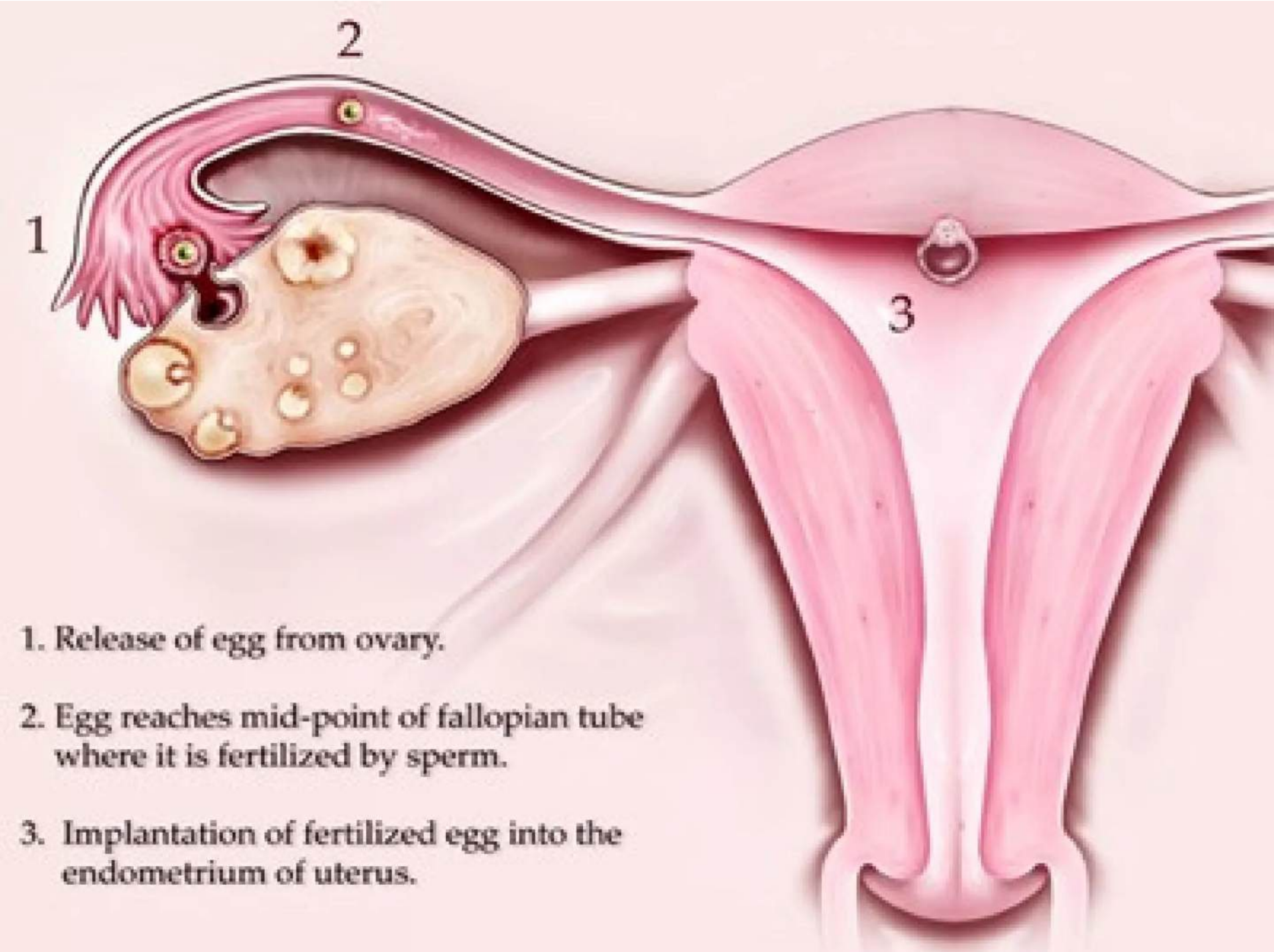


Frontal Section of Female Reproductive System

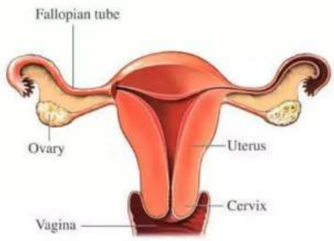


Fallopian tubes [uterine tubes]

- Stretch from the uterus to the ovaries and measure about 8 to 13 cm in length.
- The ends of the fallopian tubes lying next to the ovaries feather into ends called fimbria
- Millions of tiny hair-like cilia line the fimbria and interior of the fallopian tubes.
- The cilia beat in waves hundreds of times a second catching the egg at ovulation and moving it through the tube to the uterine cavity.
- Fertilization typically occurs in the fallopian tube


2
1
3
1. Release of egg from ovary.
2. Egg reaches mid-point of fallopian tube where it is fertilized by sperm.
3. Implantation of fertilized egg into the endometrium of uterus.

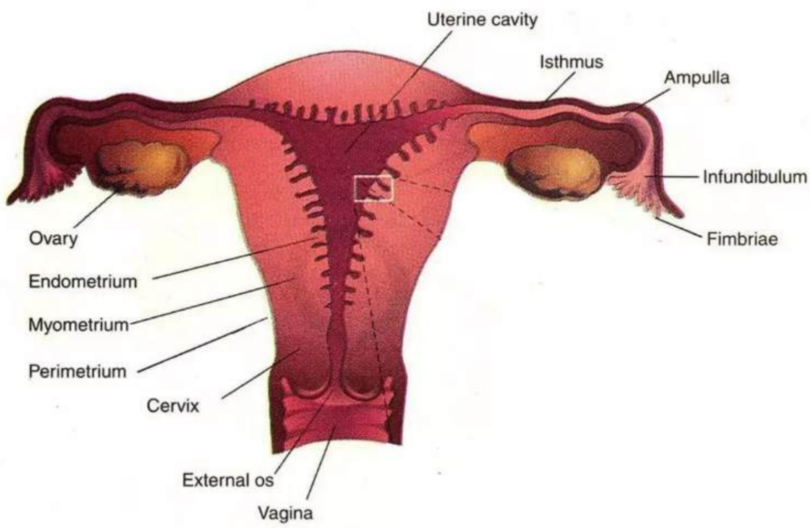
Uterus



- Pear-shaped muscular organ in the female reproductive tract.
- The fundus is the upper portion of the uterus where pregnancy occurs.
- The cervix is the lower portion of the uterus that connects with the vagina and serves as a sphincter to keep the uterus closed during pregnancy until it is time to deliver a baby.
- The uterus expands considerably during the reproductive process.
- The organ grows to from 10 to 20 times its normal size during pregnancy.

Uterus

- The main body consists of a firm outer coat of muscle (myometrium) and an inner lining of vascular, glandular material (endometrium).

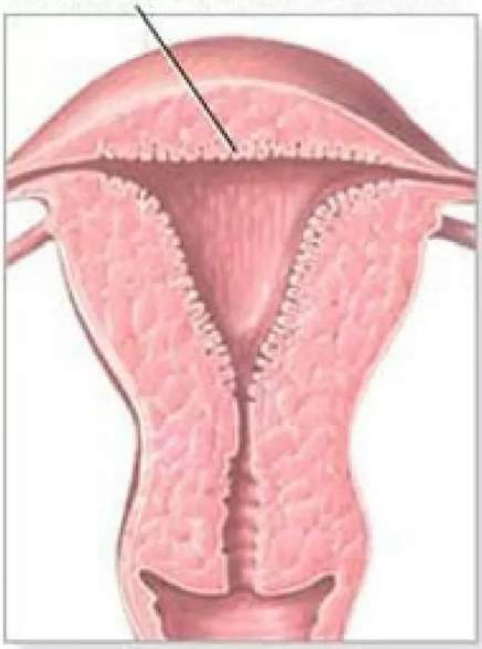


- The endometrium thickens during the menstrual cycle to allow implantation of a fertilized egg.

- Pregnancy occurs when the fertilized egg implants successfully into the endometrial lining.

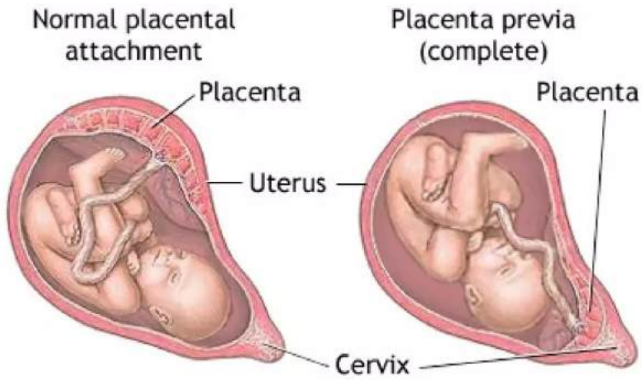
Endometrium

- The endometrium is the innermost layer as a lining for the uterus
- During the menstrual cycle, the endometrium grows to a thick, blood vessel-rich, glandular tissue layer.
- This represents an optimal environment for the implantation of a blastocyst upon its arrival in the uterus.



Endometrium

- The endometrium is central, echogenic (detectable using ultrasound scanners), and has an average thickness of 6.7 mm.
- During pregnancy, the blood vessels in the endometrium further increase in size and number, forming the **placenta**,
- Placenta supplies oxygen and nutrition to the embryo & fetus.



Presentation by

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NORMAL MENSTRUAL CYCLE

mean duration of the MC

Mean 28 days (only 15% of)

Range 21-35

average duration of menses

3-8 days

normal estimated blood loss

Approximately 30 ml

ovulation occur

Usually day 14

36 hrs after the onset of mid-cycle LH surge

the phases of the MC & ovulation regulates by:

Interaction between hypothalamus, pituitary & ovaries

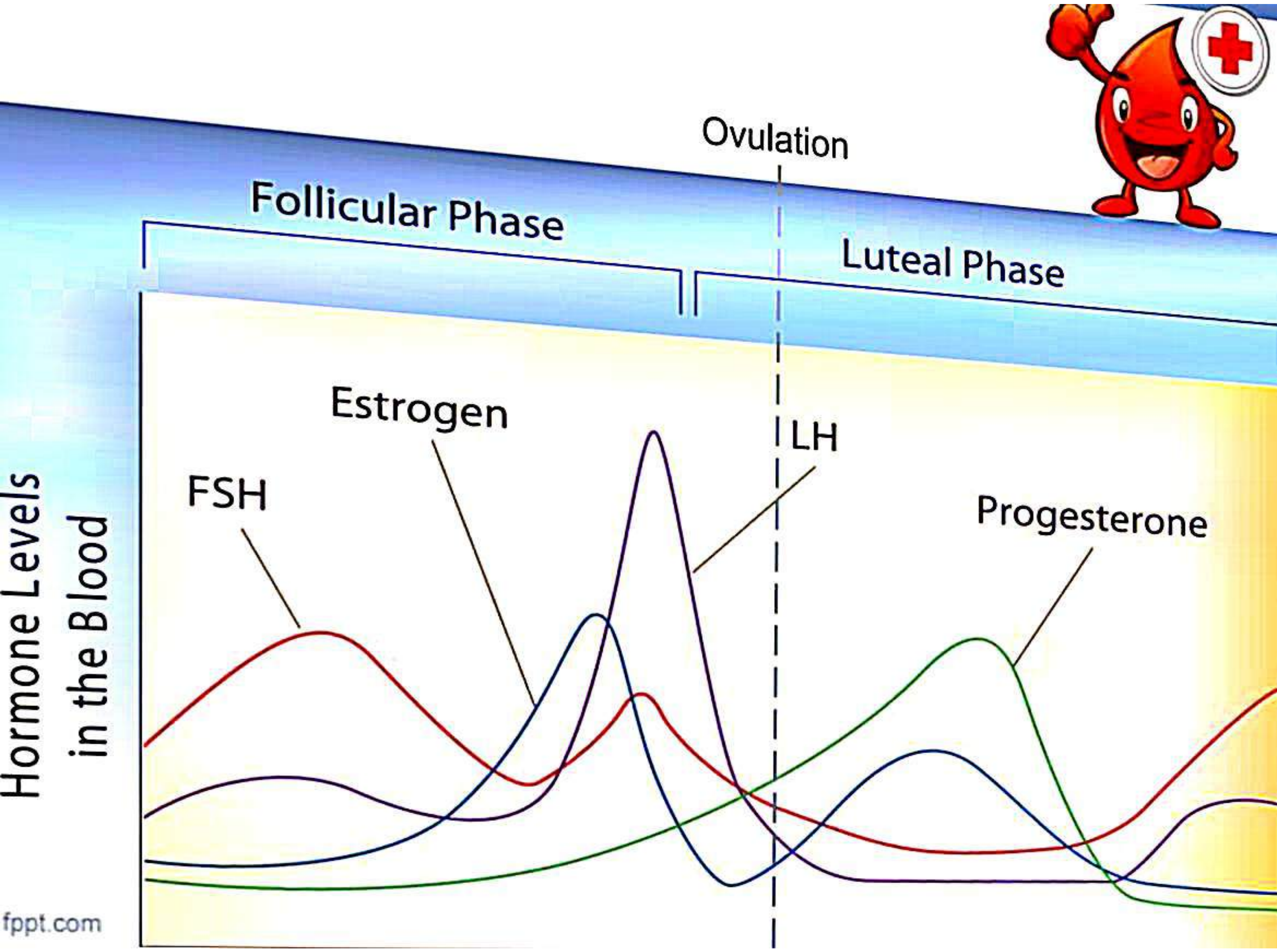
mean age of menarche & menopause are:

Menarche 12.7

Menopause 51.4

The Cycle

- Strongly linked to the endocrine system (hormone based and paracrine based)
- Typically takes 28 days to cycle through 4 phases
  - Follicular
  - Ovulation
  - Luteal
  - Menstruation
- Hormones raise and fall



Ovulation
Follicular Phase
Luteal Phase
Estrogen
LH
FSH
Progesterone
Hormone Levels
in the Blood
fppt.com

Follicular

- Begins when estrogen levels are low
- Anterior pituitary secretes FSH and LH, stimulation follicle to develop
- Cells around egg enlarge, releasing estrogen
- This causes this uterine lining to thicken

Ovulation

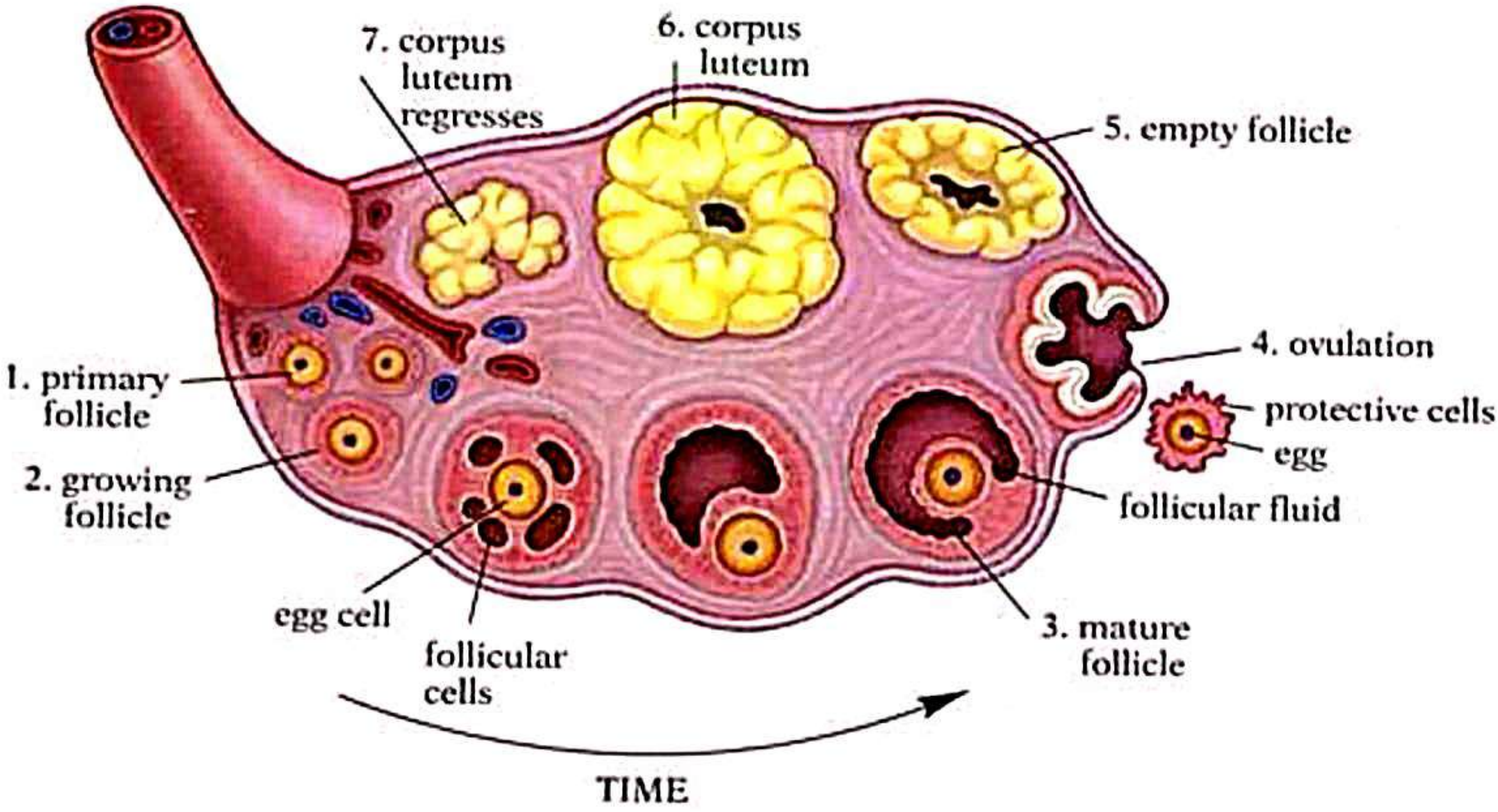
- LH and FSH still being released, for another 3-4 days
- Follicle ruptures, releasing ova into the Fallopian tubes

Luteal

- Now empty follicle changes to a yellow colour, becomes corpus luteum
- Continues to secrete estrogen, but now beings to release progesterone
- Progesterone further develops uterine lining
- If pregnant, embryo will release hormones to preserve corpus luteum

Menstruation

- Menstruation
- If no embryo, the corpus luteum begins to disintegrate
- Progesterone levels drop, uterine lining detaches, menstruation can begin
- Tissue, blood, unfertilized egg all discharged
- Can take from 3-7 days



7. corpus luteum regresses
6. corpus luteum
5. empty follicle
4. ovulation
protective cells
egg
follicular fluid
1. primary follicle
2. growing follicle
egg cell
follicular cells
3. mature follicle
TIME

PHYSIOLOGYOF THE MENSTRUAL CYCLE

Ovulation divides the MC into two phases:

1-FOLLICULAR PHASE

-Begins with menses on day 1 of the menstrual cycle
& ends with ovulation

RECRUITMENT

FSH ⇨ maturation of a cohort of ovarian follicles "recruitment"
⇨ only one reaches maturity

FOLLICULAR PHASE

MATURATION OF THE FOLLICLE (FOLLICULOGENESIS)

FSH ⇨ primordial follicle
(oocyte arrested in the diplotene stage of the 1st meiotic division surrounded by a single layer of granulosa cells)

⇨ ⇨ Primary follicle
(oocyte surrounded by a single layer of granulosa cells basement membrane & thica cells)

⇨ ⇨ Secondary follicle or preantral follicle
(oocyte surrounded by zona pellucida , several layers of granulosa cells & theca cells)

FOLLICULOGENESIS (2)

⇨⇨ tertiary or antral follicle

secondary follicle accumulate fluid in a cavity "antrum"

oocyte is in eccentric position

surrounded by granulosa cells "cumulous oophorus"

ENDOMETRIAL CHANGES DURING THE MENSTRUAL CYCLE

1-Follicular /proliferative phase

Estrogen ⇨ mitotic activity in the glands & stroma ⇨
↑ enometrial thickness from 2 to 8 mm
(from basalis to opposed basalis layer)

2-Luteal /secretory phase

Progestrone ⇨- Mitotic activity is severely restricted
-Endometrial glands produce then secrete glycogen rich vacoules
-Stromal edema
-Stromal cells enlargement
-Spiral arterioles develop, lengthen & coil

MENSTRUATION

- Periodic desquamation of the endometrium
- The external hallmark of the menstrual cycle
- Just before menses the endometrium is infiltrated with leucocytes
- Prostaglandins are maximal in the endometrium just before menses
- Prostaglandins ⇨ constriction of the spiral arterioles ⇨ischemia & desquamation

  Followed by arteriolar relaxation, bleeding & tissue breakdown

HYPOTHALAMIC ROLE IN THE MENSTRUAL CYCLE

- The hypothalamus secretes GnRH in a pulsatile fashion
- GnRH activity is first evident at puberty
- Follicular phase GnRH pulses occur hourly
- Luteal phase GnRH pulses occur every 90 minutes
- Loss of pulsatility ⇨down regulation of pituitary receptors ⇨ ↓ secretion of gonadotropins
- Release of GnRH is modulated by –ve feedback by:
  - steroids
  - gonadotropins
- Release of GnRH is modulated by external neural signals

THANKYOU

Education's purpose is to replace an empty mind with an open one"
~ Malcolm Forbes

fpp